

FIT9137

Semester 1 2025

Assignment 3 Specification

Submission Guidelines:

- **Deadline:** Semester-1 Week-14 Tuesday 2025 [10th June 2025 11:55 PM]
- You need to submit
 - **a report** (a PDF file including your screenshots with file name format as FirstName_STUDENT_ID.pdf), and
 - **your network configuration** file (the CORE file name format as FirstName_STUDENT_ID.imn) containing the required changes to complete the assignment tasks.
- Both files must be submitted via Moodle under a single submission link.
- Do **not** submit a compression of multiple files. Such submissions may receive a mark of zero.
- A handwritten document is **not** acceptable and will **not** be marked even if converted and submitted electronically.
- It is the student's responsibility to make sure that the submitted files can be opened on a standard computer (without requiring specialised software), and that all contents such as images and texts shown are understandable/readable (in English). After uploading the files as **draft** (before **finalising** the submission), **we recommend you download your submitted files and check that they open and run properly. If the files are not readable, openable, or corrupted, then you may receive a mark of zero.** Once you finalise your submission, you will not be able to revise it.
- Assignment files left in **draft** mode are **not** accepted and will **not** be marked. Make sure to finalise your submission by the deadline.
- Written texts in your PDF must be submitted as actual texts, and **not** an image of a text. Accordingly, screenshots of typed texts as images are **not** accepted (of course, this rule excludes the screenshots you take to show the computer's display).
- For the CORE configuration file, you may see a warning/error message returned by Turnitin after you upload your files. That is completely fine and **you can simply ignore the Turnitin warning/error message for the IMN file.**

- All special consideration/extension requests must be submitted via <https://www.monash.edu/students/admin/assessments/cant-complete>

Marks: This assignment is marked out of **100** marks, and it is worth **45%** of your UNIT total marks. You must submit both the CORE configuration file and the PDF report. If you submit only one of the files, you will receive a mark of zero.

Feedback: Your Tutor will provide you with marks and feedback (if your submitted PDF file and the CORE *.imn file are readable and markable)

Brief Description

Assignment will include the materials covered within Weeks 6-12. In particular, the assignment will consist of questions related to network and transport layers, structures and functions of local area, backbone and wide area networks, and network security. The format of the student submission will be a written report and a network configuration.

Learning Outcomes covered

This Assignment covers material from Weeks 6-12. This is an **individual** assignment. By completing this Assignment, you will understand the learning outcomes 4 to 7.

- Examine networks using the underlying fundamental theories, models and protocols for data transmission.
- Analyse and formulate the functions and architectures of (wireless) local area networks, wide area networks and the Internet.
- Identify cybersecurity threats and ethical considerations on the Internet.
- Apply and implement cybersecurity enabling techniques and countermeasures such as virtual private networks (VPN).

Academic Integrity

This is an **individual** assignment and group work is prohibited. It is an academic integrity requirement that your submitted work be original. Penalties will be applied to the whole submission if there is any evidence of copying, collaboration, pasting from websites, or copying from textbooks. When asked to use the Internet, books, or other academic resources to answer a question, it does not mean to copy the text verbatim from the source. You must write the answers in your own words such that your understanding of the answer is evident. You must always cite your references within the text and list them at the end of the report. Academic integrity policies apply to all assessments. You can refer to this link below:

<https://www.monash.edu/student-academic-success/learnhq/maintain-academic-integrity>

Penalties

Late submissions will result in a 5% deduction of the total marks per calendar day (up to 7 days). For example, if you get 80/100 marks originally and submit 2 days late, then you would get a 10-mark penalty (5 marks/day) and your final marks would be 70/100. Note that the total number of late submission days is calculated by rounding up, e.g., if a submission is strictly more than 5 days (and at most 6 days) late, then a 6-day penalty would be applied. Submissions **more** than 7 calendar days late will receive a mark of zero (0) and no assessment feedback will be provided.

Network Configuration and Security

Introduction

For this assignment you will use the **core network emulator** to complete a series of tasks on an **individual** core configuration file that is generated for you. To download your individual core configuration file, open the unit's Moodle page then navigate to the Assessments section and follow the provided instructions for Assignment 3. The downloaded ***.imn** file will be in **zip** format, please **unzip** the file to use it.

You must write a report to explain

- the changes you make,
- the configurations you add to achieve the goals of each task,
- your reasons for each change/configuration, and
- the tests you perform to check the task is accomplished.

Your submitted core file will be marked by running the configuration and testing that the tasks are completed. The report will serve as a reference to be checked during marking. However, if a test fails when running your submitted core file, you will receive no mark for that failed test (i.e., part of a task) regardless of your explanations in the report. If tasks are similar, you only need to explain your reasons once, and then just report the changes you make to individual services on each node.

Network Structure

The provided network is composed of two organisations labelled **Talos** and **Delos**, a router named **Internet** playing the role of the Internet, and a global DNS server named **clio**. The internal subnets of **talos** are labelled **Internal**, and the public servers of the **talos** network are placed in a separate subnet named **DMZ**. The Internet facing router of the Talos organisation, **R3**, is also its network firewall. The Delos network is divided into two subnets: **(i)** a subnet for the organisation clients and private servers and **(ii)** a subnet for its public servers. The public servers of **Delos** are named **apollo**, **artemis**, and **demeter** providing web, domain name, and mail services respectively.

DNS Setup

The core file is configured to resolve the domain names between the two organisations, **talos.edu** and **delos.edu**. This is achieved through a global DNS server named **clio**. The server only resolves the names for the two domains in the configuration (**talos.edu** and **delos.edu**) by sending the request to the corresponding nameserver for each domain and sending back the response to the requesting client. Each DNS server in aforementioned networks must have access to UDP port 53 of the server **clio** as the organisation DNS servers resolve the names on behalf of their respective

clients. You do not need to make any changes to DNS servers; this section only explains the DNS setup.

Important Notes

- It is recommended to use **tcpdump** if you wish to capture traffic and to observe whether the packets reach their intended destination when trying to accomplish the tasks. To use **tcpdump**, you can right click on a node and move the mouse to select tcpdump in the provided list and then select the intended interface. You can also run tcpdump from the command line using the command **tcpdump -l -i eth0** to print the summary of the captured packets from the eth0 interface in the terminal. To write the captured packets to a file, use the command with **w** option followed by a filename. For instance, running the command **tcpdump -w /home/muni/R3_eth3.pcap -i eth3** on the node **R3** will capture the traffic on its **eth3** interface and store the frames in a file named **R3_eth3.pcap** under **/home/muni** directory. You can then stop the capture with Control+C and use Wireshark to analyse the captured packets.
- Any changes you make to the nodes when the emulation is running will be lost when you stop the emulation. You can test the changes you want to make when the emulation is running and once you have the correct commands, then add them through the GUI in the proper service. For example, to add static routes to a router that persist and will be stored with the configuration file, you need to add **ip route add** commands to the **StaticRoute** service of that router.
- If you make changes to a core configuration file and then close the CORE window without saving the changes, you will **not** be warned, and **the changes will be lost**. Hence if you wish to keep the changes you have made, you must save these changes **before** closing the CORE window. Please note: changes are to be saved for static routing, DHCP configuration & Firewall configurations.
- Make sure to keep a backup of your CORE file somewhere outside of the VM in case you encounter issues with your VM, and you need to replace the VM. This will make sure that you would not lose the work you have done. It is your responsibility to back up your work.
- In the provided configuration file, you must **NOT** alter the overall network structure or topology, including device and node names or IP addresses, nor can you add or remove nodes. Additionally, you are **NOT** allowed to modify link speeds or link delays, and implementing any extra network services such as VLAN, VPN, or NAT is **prohibited**. If you do not obey these rules, your submitted work will be deemed invalid and will receive a zero mark.

Tasks

Task A: Routing

[35 + 10 = 45 Marks]

The routing tables of the routers in the provided network are **NOT** configured. The correct configuration of this task allows **any host from any network to reach any other host** in the entire network. You must satisfy the following requirements while completing static routing task:

1. All routers must be configured solely with a static routing table; dynamic routing is **NOT** allowed. Static routing should be set up using the given link delays (which include all of processing, transmission and propagation delays) specified in the provided configuration file. You don't need to consider the link speed for this part.
2. All hosts inside the **talos.edu** network must be reachable from any other host within that network through an *optimal path*. You need to add static routes to routers **R1**, **R2**, **R3**, and **R4** to accomplish this goal. You must explain your reasons for choosing a path in the report. The notation **us** for links represents the propagation delay in microseconds. You can assume that the processing and transmission delays at the routers are negligible.
3. The router **R3** must be the default gateway of the **talos.edu** network. The router **Internet** must be the default gateway of **R3** and **minerva** (the only router of Delos). You will lose marks if you create routing loops.

Task B: DHCP Server

[8 + 2 = 10 Marks]

The clients of **delos** are configured with static IP addresses. Your task is to:

1. Configure DHCP server on the node **minerva** to assign dynamic IP addresses and other required settings to the client machines in the client's subnet.
2. You can make use of the DHCP server configuration on **R1** as a reference to follow.
3. Enable DHCP client service on clients of **delos**.

Note: The node **leto** is a private local server in the client's subnet and must have a static IP address as assigned for the given configuration.

Task C: Firewall

[45 Marks]

The node **R3** is the firewall for **talos** network. Configure the Firewall service on this node to satisfy the following requirements:

1. Allow traffic from anywhere to DMZ for the provided service by each server. This must be limited to only the public service(s) that a server provides: **dns** only DNS, **web** only HTTP, **mail** only SMTP. [9 Marks]
2. Allow servers in DMZ to **initiate** a communication if it is required by the service the server provides and only for that service (stateful inspection: DMZ → External). [6 Marks]
3. Allow internal hosts to access all services provided by servers in the DMZ (stateful inspection: Internal → DMZ). This includes all services that DMZ servers provide. You can be more permissive here and use address ranges and all IP traffic. All servers in DMZ run SSH service which you can use to test your rules for the internal subnets. [6 Marks]
4. Allow internal hosts to reach other internal hosts (this is only if the internal traffic passes through R3). All traffic is allowed if it is internal to internal communication. [3 Marks]
5. Allow internal nodes to access external servers however packets from external to internal are only allowed if they are responses to communications that were initiated from inside (stateful inspection: Internal → External). [9 Marks]
6. Allow only the nodes in clients subnet of **talos** to ssh to node **R3** (any host connected to the **R1.eth0** subnet). [4 Marks]
7. Permit node R3 to send ICMP echo request messages to internal talos nodes and all DMZ servers and receive ICMP echo replies from those nodes and servers. Also, permit internal talos nodes and all DMZ servers to send ICMP echo request messages to node R3 and receive ICMP echo reply from the node R3. [8 Marks]
8. All other traffic must be dropped (see Notes below).

Important Notes for Task C:

- If the **Requirement 8 is not satisfied**, you will receive a **zero mark for the firewall task** regardless of any other correct rule you add as it would expose the entire network. You will lose partial marks if your rules are too permissive allowing more traffic than specified to reach the destination for each requirement.
- You only receive marks if the test for each requirement succeeds. No partial marks will be given if only part of a rule is correct. When two rules are required for the incoming and outgoing traffic, no partial marks will be given if one of the rules is correct.
- For stateful inspection, the traffic is allowed if it is initiated from the more trusted side of the firewall to the less trusted side. The traffic in the opposite direction, from the less trusted interface to the more trusted interface, is only allowed if the packets are the responses to an initiated communication from the more trusted side. The trust level in the requirements is indicated as **Higher** → **Lower** for each stateful inspection, meaning the connection initiation is allowed from the higher level to the lower level and only the responses for the initiated connections are allowed from the lower level to the higher level.

- If you have reachability issues in Task A, i.e. a host is not reachable from another host, you may lose marks in firewall tests as well when the traffic must be allowed. You will not lose any marks for firewall rules if a host is reachable but through a sub-optimal path and the firewall rules are correct.
- **You must submit the CORE file with Firewall service enabled on node R3. The service is enabled in the individual files without any rules hence all traffic is allowed.**

Reference Notes

Please acknowledge any reference appropriately and the use of any AI tools (refer to the assignment link on Moodle to access information from Learn HQ).